

SEDAR 9: Gulf of Mexico Gray Triggerfish, Greater Amberjack, & Vermilion Snapper

Gray Triggerfish

Data Workshop

No research recommendations were provided.

Assessment Workshop

No research recommendations were provided.

Review Workshop

- The Review Panel should be provided an executive summary for substantive documents from Data and Assessment Workshops, a succinct table of model structural equation and parameters, and if appropriate a table of management options. A glossary of all the acronyms used in the assessments should be provided as an appendix in every assessment report.
- All of the data used for the assessment should be included in the Reports as well as the model formulations for the assessment. Some of the data in gray Triggerfish (such as age composition data) used in the assessment were missing from the Assessment Report, which could preclude further independent evaluation of the assessment results. The Addendum to the gray triggerfish Assessment Report includes these data now.
- An observer program should be implemented to estimate levels of shrimp bycatch and appropriate age composition with some well-designed, systematic research programs, which are essential to provide the data necessary for effective management. Shrimp bycatches for gray triggerfish are the dominant removals for this species and it is scientifically important for better estimates for an accurate stock assessment. Catch in numbers of fish is dominated by shrimp bycatch which mainly consists of age-0 and age-1 fish (Table 1 and Fig 1 in the Addendum). The shrimp bycatch fishery annually removes roughly 1 million fish age-1 equivalent and peaked at 5 million fish at year 2002. However the recreational and commercial fisheries' combined take was roughly 1 million pounds in recent years but had past peaks reaching 3 million pounds annually.
- A comprehensive age-reading programme should be established in the major sectors. This will allow a more accurate age distribution and therefore a more accurate and precise assessment. This is more important for this species since the assessment method has changed from ASPIC model to SSASPM using catch at age data.
- MRFSS programme should be strengthened so that more precise estimations of total catches are available for the assessment.

- A mark-recapture study should be initiated. Such a study will help:
 - * Identifying movements and migrations between east and west regions;
 - * Estimating fishing mortality;
 - * Enhancing the population estimates; and
 - * Identifying the stock structure;
 - * Better understanding habitat preferences.
- The methods should be more thoroughly documented, including the structural model equations, the observation-error models, process-error models (if appropriate), values of constants, constraints and priors, and description of the fitting algorithm including the uncertainty-estimation method.
- The panel should be provided more detailed model diagnostics, such as complete lists of estimated parameters together with their estimated standard errors, the most important investigation of model sensitivity runs.
- The model residuals diagnostics should be included to test whether there is still time-series autocorrelation for lack of goodness of fit in the assessment.
- The resources available to the assessment data collection, processing and modeling teams should be significantly increased. This increase in resources would be required in order to allow the foregoing recommendations to be implemented realistically.
- The panel's internally-adopted guidelines for assessing assessments developed during the SEDAR 9 Review Workshop (see Appendix 1) should be followed.

Greater Amberjack

Data Workshop

No research recommendations provided.

Assessment Workshop

- age-length keys representative of all sectors and regions of the fishery in the U.S. Gulf of Mexico (in part being addressed by current MARFIN NA05NMF4331071).
- reproductive parameters, such as age of sexual maturity and fecundity at age for the Gulf of Mexico stock of amberjack (age at maturity being addressed by current MARFIN NA05NMF4331071).
- fishery-specific release mortality

Review Workshop

- collect information on the species composition and total catch of shore based landings of Greater Amberjack and other species.
- Within the greater amberjack assessment, because of the uncertainty caused by the final year of data, an update assessment should be conducted within a few years (outside the usual benchmark assessment process) to elucidate the most likely trajectory being followed by the stock and enable the provision of remedial management measures should these be necessary.
- A yield-per-recruit analysis should be made for the greater amberjack as an addition to future assessments to act as a check against growth overfishing and to determine whether the legal minimum length is appropriate.

Vermilion Snapper*Data Workshop*

No research recommendations provided.

Assessment Workshop

No research recommendations provided.

Review Workshop

- Establish an obligatory, randomised observer scheme to estimate levels of shrimp by-catches.
- Establish a comprehensive age-reading programme for vermilion snapper in the major sectors, especially the shrimp by-catches.
- Consider further reinforcing the MRFSS programme so that more precise and accurate estimations of recreational catches can be obtained.
- Methods should preferably be simulation-tested prior to their use in an advisory context.
- Methods should be documented more fully, including the structural model equations, the observation-error models, process-error models (if appropriate), values of constants, constraints and priors, and description of the fitting algorithm including the uncertainty-estimation method. This documentation, together with the input data, should be included in the stock assessment reports.
- More detailed model diagnostics should be provided, such as complete lists of estimated parameters together with their estimated standard errors.

- Significant increases in the resources available to the data collection, processing and modeling teams would be required in order to allow the foregoing recommendations to be implemented.
- The benchmarks should be updated when new life history parameters become available.
- In future assessments the SSASPM should be modified to take account of bias-correction in the length-weight prediction.

General SEDAR Process Recommendations

Specific Recommendations of the Review Workshop Panel

- There were some concerns expressed in the Review Workshop that pressure may have been brought to participants at some of those workshops to progress management further than was possible within the available time frame and with available time series data.
- Incorporation of fishermen's knowledge into the data and assessment process.
- Whenever a major data stream (effort, catches or catch rates) is to be modified the details of any modifications should be stated explicitly and documented completely.
- To avoid overloading the scientific staff, sufficient resources and time should always be provided to prepare the materials to normal scientific standards and allowance be made for any major un-avoidable disruption to this process (such as Hurricane Katrina).
- A summary table for each assessment should be provided stating each data stream to be used with its constraints and any treatments or modifications made. Included in this table should be an indication of the reliability of each data stream. It could be included in either the Data Workshop or Assessment Workshop reports.
- Each assessment document should, preferably, contain appendices detailing the structure and likelihood estimator for at least the base case model, or alternatively refer to a readily available document containing these details.
- The various model outputs and management benchmarks (e.g. MSY, Fmsy, Bmsy, MSST, MFMT) for the accepted base case model should be defined in one place within the stock assessment report along with how they were defined mathematically.
- A glossary of all the acronyms used in the assessments should be provided as an appendix in every assessment report.
- If the data available are adequate for conducting an assessment, then the 5th and 6th Terms of Reference in the Data Workshop should be removed from consideration by the Data Workshop and shifted instead to the Assessment Workshop.

- There was large volume of documentation associated with this Review Workshop. The Review Panel recommended the need for a clear executive summary for all substantive Data and Assessment Documents. It could be more informative to distribute a succinct table of model equations and parameters (estimated and observed) to be provided for each assessment along with, if appropriate, a table of management options (e.g. a decision table) and the risks associated with them.
- The SEDAR process appears to be remarkably thorough and detailed, with many opportunities for clarification and communication of the stock assessment processes. The whole idea of such detailed reviews is to be applauded as demonstrating a willingness to be open and to provide the best defensible assessments possible with available data.
- The process itself is relatively intensive and after observing the difficulties involved in review three species at the same time it is recommended that future SEDAR events only consider two species at the most. With three fisheries there are greater opportunities for confusion between species and the time available for detailed discussion could be compromised. If there were to be multiple species considered in future SEDAR workshops it would be beneficial to allocate species among reviewers prior to arrival at the workshop so they could begin the detailed and focused examination of the very many reports from the Data and Assessment Workshops before arriving at the review venue.
- The final review workshop report appears to be asking for the review panelists to produce an independent assessment summary and while the review panel may have possibly provided significant input to the assessment development the work is still mostly all that of the assessment scientists. As such it feels contrary to general practice to not have their names associated with the final consensus report.
- Some of the review reporting, such as the advisory report, appears to be primarily an editorial effort which could be produced by anyone rather than the review panelist. The chances for errors of omission would be significantly lower if the advisory report were produced by the assessment scientists concerned and merely edited and agreed to by the review panelists.

Recommend Approach to Assessment Review

- The review panel considered the characteristics that would ideally be desirable in a stock assessment process used for advisory purposes.
 1. All relevant data should be used, unless there is an *a priori* reason to exclude a data series, or a sound *a posteriori* reason can be identified. Data should be real observations, not “filled-in” using assumptions or other criteria, to the extent possible. Fish stock assessment depends on having reasonably long time-series of catch, effort and fishery-independent abundance estimates.

2. Conclusions about stock status with respect to reference points should be robust to underlying assumptions about data and structural model, e.g. reliance on filling-in assumptions, dependence on most contested parts of the data sets.

3. Assessments should include the following :

- Data screening, to check assumptions in 1 and 2.
- Model screening, to see if broadly similar conclusions are drawn from different models, including sensitivity to constraints etc.
- Residual pattern screening: Does the model replicate the trends in the data?
- Credibility check: are the estimated model parameters reasonable (e.g. selection pattern, r , B_0/B_{msy} , trends in F etc. in the context of biological knowledge about the stock and the fishery ?
- Variance estimates (or posteriors) for the estimated interest parameters, and a priori model testing, using simulated data, which should demonstrate that the model has useful precision in predicting interest parameters when presented with data.

4. Assessment documentation should include :

- Data used to fit the assessment model.
- Structural model equations, including process-error model if applicable
- Observation-error model
- Description of estimating algorithm
- List of final parameter estimates and their sd.s
- Computational validation, including simulation testing
- Source code (and ideally documentation) of the programs used should be made available.

Recommendations of the CIE Contractors

- Whenever a major data stream (effort, catches or catch rates) is to be modified the details of any modifications should be stated explicitly and documented completely.

- To avoid overloading the scientific staff, sufficient resources and time should always be provided to prepare the materials to normal scientific standards and allowance be made for any major un-avoidable disruption to this process (such as Hurricane Katrina).
- A summary table for each assessment should be provided stating each data stream to be used with its constraints and any treatments or modifications made. Included in this table should be an indication of the reliability of each data stream. It could be included in either the Data Workshop or Assessment Workshop reports.
- Each assessment document should, preferably, contain appendices detailing the structure and likelihood estimator for at least the base case model, or alternatively refer to a readily available document containing these details.
- The various model outputs and management benchmarks (e.g. MSY, Fmsy, Bmsy, MSST, MFMT) for the accepted base case model should be defined in one place within the stock assessment report along with how they were defined mathematically.
- A glossary of all the acronyms used in the assessments should be provided as an appendix in every assessment report.
- The SEDAR process is impressive in its thoroughness, its transparency, and in the consensus perception of stock development that it builds. This consensus-building is however achieved at considerable cost in terms of scientific manpower. The three-stage process of data evaluation, stock assessment and review is laudable in principle, but each stage involves a large number of participants, many of which are to some extent repeating work that has been done elsewhere. A symptom of this is that the technical elements of the assessments are spread out through a large number of working documents and workshop reports which refer to each other, creating a “thicket” of documentation that is difficult for an outsider to this process to penetrate. The task of repeating text from one report to another detracts significantly from the time available to address new substance.
- The consensus-building is achieved at cost of considerable inefficiency in the use of scientific resources, to an extent that may not be sustainable.
- I would suggest that SEDAR consider some of the following options, in order of priority:
 1. Recruiting more assessment scientists to the process;
 2. Reducing and simplifying the terms of reference to workshops - in particular, it is unrealistic to expect experts in fish stock modelling to address terms of reference concerning control and enforcement issues;

3. Reducing and simplifying the number of reports to be produced – for example, there is considerable redundancy and repetition in the six reports generated by the review process;
 4. Merging some meetings in the process, e.g. either merge the “data” and “assessment” workshops into one, or else merge “data” workshops for several species (because many data issues are not species specific), or incorporate external experts into the assessment workshops and cease holding separate “review” meetings;
 5. Introducing a “lighter” procedure for assessing species of minor importance, with perhaps all three steps addressed in a single meeting.
- With respect to the SEDAR Review process in particular, I would make the following points:
 - * The workload for the reviewers to address the terms of reference thoroughly is very challenging to meet within the allocated 12 working days – this could be alleviated with some pre-meeting task allocation and possibly a stronger focus by each reviewer to a particular stock;
 - * If an agreement could be reached on the desirable elements of an assessment (e.g. as Section 2) this could assist a better coordination of the assessment and review activities.